

LabBench: Benchmarking AI wet lab experimental design and decision-making for AI Agents

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Abstract

AI agents are increasingly deployed in scenarios where expert biological reasoning is expected. We present LabBench, a benchmark of 20 held-out tasks constructed from the primary records of real-world wet labs, a drug discovery program and an academic genomics lab, and graded against binary criteria tied to the labs’ own recorded decisions. The evaluation demonstrates where AI agents come up short in realistic scientific reasoning: while excelling at interpreting previous experiments, often identifying correct conclusions and declining overclaims from the presented lab data, agents are routinely unable to match human-level performance in determining the specific next experiments to be conducted. The strongest agents, GPT-6 Astra and Claude Opus 5.5, score 41%; criteria that require choosing, committing or ranking pass 21% of judgments against 47% for identifying what a measurement is, and no agent passed any of the 13 criteria on which experiment should come first. A common issue in existing benchmarks is unreasonable expectations or criteria for the agent to complete, making them effectively impossible. We test against this directly in LabBench, to verify that agents can fairly complete these tasks. By appending a one-sentence hint that points in the correct direction without stating the answer, we ablate GPT-6 Astra on five core decisions that no agent reached, and demonstrate that it then reaches the reasoning for all five, indicating that the requisite knowledge is latent but not recalled unprompted.

1 Introduction

Accelerating biological experimentation relies on the ability to quickly make expert-level decisions about which experiments are needed next. Given a set of results, choosing the next test that can most efficiently falsify or strengthen the working hypothesis is a critical skill that currently can only be reliably performed by human experts, and a poorly chosen experiment can cost a lab weeks of bench time and reagents. To scale the decision-making behind these discoveries past human capacity, we must evaluate and improve AI agents along this axis. LabBench v1 spans seven experimental domains: nascent transcription, gene regulation, 3D genome organization, assay and library development, target and disease-model validation, hit-to-lead chemistry, and drug-response profiling.

We construct the benchmark from the historical artifacts of real-world wet-lab work, spanning drug discovery and academic genomics research. Each task is a snapshot in time: the agent receives the experimental records that existed at a decision point (files include slide exports, figures, tables, scripts and sequencing outputs), while the lab’s own interpretation and subsequent decision are withheld. The agent must produce a report that commits to a decision: in six tasks, a ranked set of next experiments; in the rest, which condition, model, comparison or claim the evidence supports.

Ground truth is the decision the lab’s scientists actually made. Each task is graded against 20–22 binary criteria, each tied to the visible evidence and the hidden record that supports it, so an agent is scored both on whether it reaches the experts’ decision and on the reasoning that leads there.

Our core finding is that agents are often extensively informed about the specific topic, but routinely fail to connect the evidence to the correct next step. Across five frontier agents, 45% of criteria were passed by none, including 9 of the 23 core decisions, and in the six tasks that ask for ranked next experiments, no agent passed any of the 13 criteria on which experiment should come first. With a single sentence added to the prompt that directs the agent toward the relevant evidence without stating the answer, GPT-6 Astra reached all five never-passed core decisions we tested, reasoning about and designing experiments comparable to the work of human experts (Figure 4). The knowledge is latent, but agents do not recall it unprompted.

2 Related work

Scientific knowledge and question answering. Early benchmarks measure what a model knows: broad academic knowledge [1], multimodal science questions with explanations [2], graduate-level questions written to resist lookup [3], and expert questions at the frontier of each field [4]. These benchmarks measure whether a model holds scientific knowledge. LabBench measures whether it applies that knowledge, unprompted, to evidence it has not seen: each answer is a decision recorded in a lab’s own archive rather than a fact in the literature. The two diverge: agents that demonstrably hold the knowledge a decision needs still miss the decision (§4). We draw the tasks from internal lab records that are unlikely to be recallable from the open internet, and we expect the agent to navigate the scientific evidence as a real researcher would, without the task description pointing to the evidence that matters.

Research skills in biology. LAB-Bench tests practical research tasks such as literature and database lookup, figure and table interpretation, sequence manipulation and protocol troubleshooting [5], and BioPlanner tests the ordering of protocol steps [6]. These isolate component skills in multiple-choice or structured form. LabBench asks for the decision those skills serve, in free-form deliverables.

Agents on data analysis and reproduction. ScienceAgentBench [7], DiscoveryBench [8] and BixBench [9] evaluate agents that analyze data to reach findings; CORE-Bench [10] and SciCode [11] test reproducing computations and writing scientific code. LabBench builds on this setting but grades the interpretation and the committed next step, not only the computation: several of our tasks were solved computationally by every agent and still failed on the reasoning (§4).

Rubric grading of open-ended work. PaperBench grades research replication against hierarchical expert rubrics [12], HealthBench grades clinical conversations against physician-written criteria [13], and Harvey LAB builds legal-agent tasks from real client matters and grades the work product against atomic, binary pass/fail criteria [14]; all can be applied by a model judge [15]. LabBench’s criteria are tied to hidden primary records, and every verdict must cite the submission, the input and the human evidence.

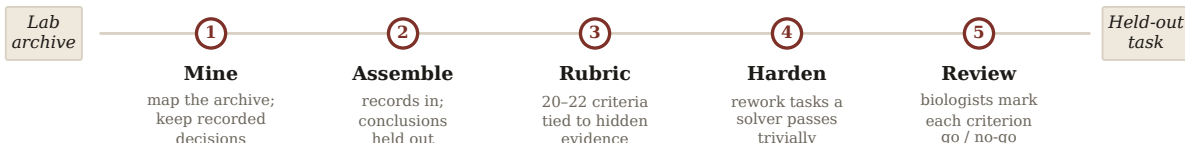
Eliciting latent capability. Prompting can reveal capabilities a model does not show by default [16, 17], which is why dangerous-capability evaluations distinguish what a model does from what it can be elicited to do [18]; models can also be pulled off course by salient but irrelevant context

[19]. Our hint probe applies this idea to scientific decisions: a sentence that redirects attention, without supplying the answer, separates missing knowledge from knowledge that is not brought to bear.

3 The benchmark

Construction. We construct these cases from our proprietary data, collected over years of wet-lab experimentation. For each scenario we reconstruct a time series of the relevant experiments and events along the path to discovery. We then identify the point at which a core decision was made and withhold its outcome, and the lab’s interpretation leading to it, from the packet of experimental records the agent receives (Figure 1A). A task prompt asks the agent for the next-step decision the lab’s expert biologists faced at that point, and 20–22 binary (pass/fail) criteria are derived from the true answer, following Harvey LAB [14], each tied to the visible evidence and the hidden record that supports it. Unlike atomic checks, criteria can depend on one another: some pass only if an earlier conclusion they build on is correct. The criteria are bootstrapped by AI agents with visibility of the input packet, the prompt and the correct answer. We then test a solver agent on each task. If it reaches the answer trivially, an AI agent inspects its trace for answer leakage and the task is hardened. The result is tasks that are difficult for frontier agents yet whose core decisions remain within reach (§4, Figure 4). Finally, expert biologists review each case packet in its entirety, checking not only clarity and feasibility but also direct relevance to the field. All 20 final tasks were manually verified by expert biologists.

A From lab archive to graded task



B Criteria by reasoning skill and theme (dot area = count)

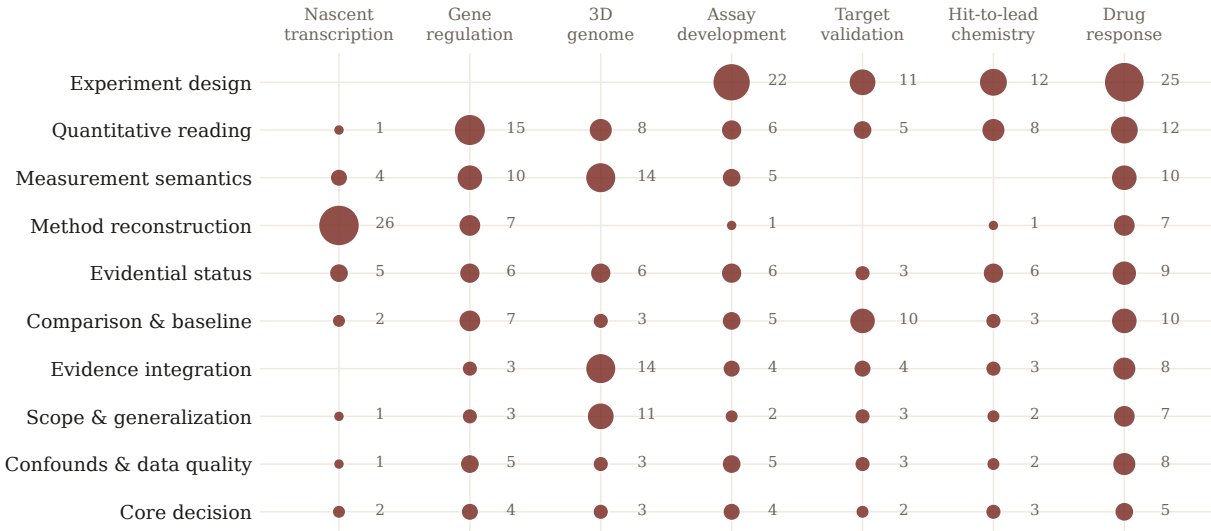


Figure 1: Benchmark design. **A**, From lab archive to graded task. **B**, Criteria per reasoning skill and theme (dot area = count).

Scoring. A fresh judge sees the brief, the documents, the hidden records and the submission, but not the agent’s trace. It judges every criterion once; a missing deliverable fails its criteria, and each verdict must cite the submission, the input and the human evidence. A task’s score is its share of criteria passed.

Coverage. The tasks span seven themes (Table 1) and 13 analyses, 4 reviews and 3 experiment designs. We labelled each criterion with the reasoning step whose failure would fail it, giving ten skills (Table 2). Six skills appear in all seven themes (Figure 1B): baseline and denominator reasoning, for instance, recurs in transcription-factor peak fractions (m6-004), motif enrichment against expected counts (m6-002) and the reference for gene-expression changes (m10-003).

Table 1: Biological themes.

Theme	Tasks	Criteria	Example task
Nascent transcription	2	42	Reconstruct how control genomic regions were chosen in a nascent-RNA analysis (m5-002)
Gene regulation	3	60	Compare the chromatin context of early- and late-bound transcription-factor sites (m6-004)
3D genome	3	62	Assess a 3D chromosome model against imaging distance measurements (m1-012)
Assay & library development	3	60	Choose library preparation conditions from fragment-size traces (t07)
Target & disease-model validation	2	41	Choose a matched cell comparison before testing a DNA-repair vulnerability (m10-007)
Hit-to-lead chemistry	2	40	Reassess a compound series and its lead after new data (t15)
Drug-response profiling	5	101	Separate plate artifacts from drug effects in a transcriptomic screen (t09)

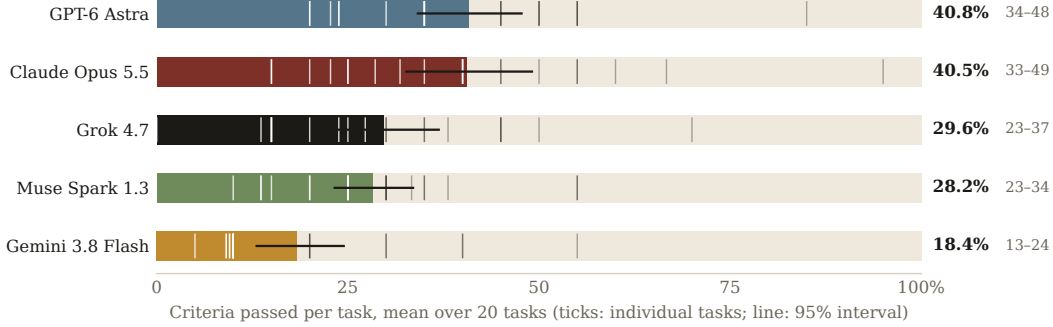
Table 2: Reasoning skills. Every criterion carries one primary skill.

Skill	What a passing answer does	Criteria
Core decision	Commits to the lab’s choice and defends it against the strongest alternative	23
Comparison & baseline	Chooses the contrast, control, denominator or normalization that answers the question	40
Method reconstruction	Recovers what an analysis did from its code and outputs, including edge cases	42
Quantitative reading	Reads values, directions and magnitudes correctly from figures and tables	55
Measurement semantics	States what a score, assay or statistic measures and what it does not	43
Confounds & data quality	Detects batch effects, failed libraries, non-independent replicates and confounds	27
Evidence integration	Reconciles conflicting or complementary results into one account	36
Evidential status	Keeps association apart from causation, and observations apart from predictions	41
Scope & generalization	Bounds claims to the system, dose, time and resolution sampled	29
Experiment design	Chooses and orders the next experiment, with controls and readouts	70

4 Results

The best agents pass two in five criteria. Each agent ran once per task in its vendor’s harness. GPT-6 Astra and Claude Opus 5.5 are tied (Figure 2A), and Astra got there in a median 5 minutes per task against Opus’s 35 (Figure 2B).

A



B

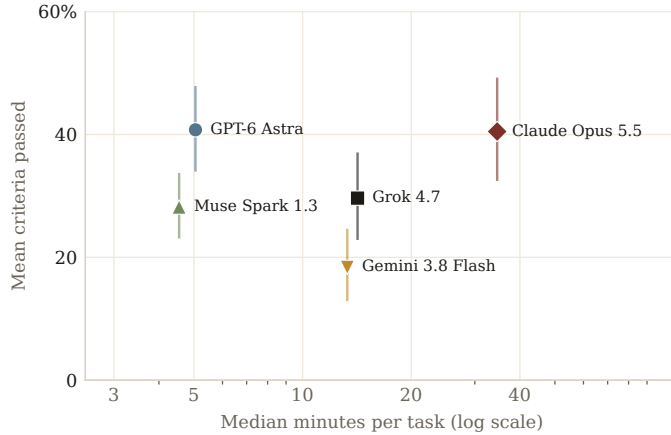


Figure 2: Scores. **A**, Mean share of criteria passed over 20 tasks; ticks are single tasks and the dark line the 95% bootstrap interval. **B**, Median solve time against mean score (log time axis).

Agents fail the same criteria. 182 of 406 criteria were passed by no agent and only 31 by all five (Figure 3A). Agents agree on 70–84% of verdicts, well above chance (Cohen’s κ 0.30–0.67 [20]). What one frontier agent misses, the others usually miss too.

They identify and qualify; they rarely choose. Criteria passed by four or five agents ask for single, local steps the records make explicit: saying what a measurement is (a modeled cluster’s size is not a cell frequency; **m1-005**), declining an overclaim (a predicted drug response is not a completed validation; **m8-003**), or reconstructing what an analysis did (applying a reported identity correction to the right experiment; **t15**). Criteria that ask the agent to choose, commit or rank pass 21% of judgments, against 47% for identifying what something is (Figure 3B). Nine of 23 core decisions were never passed, and in the six briefs that ask for ranked next experiments, no agent passed any of the 13 criteria about which experiment comes first. The other common misses are defending a choice against its alternative, reporting every item rather than a summary, walking through edge cases, reconciling several lines of evidence, and the precise meaning of derived quantities.

A

Agents passing the criterion, of five. Cells give criterion counts; shade is the share of the row.

	0 (none)	1	2	3	4	5 (all)	
All criteria	182	55	40	45	53	31	n=406
Method reconstruction	17	4	4	3	6	8	n=42
Measurement semantics	12	9	6	5	5	6	n=43
Evidential status	14	6	4	5	9	3	n=41
Comparison & baseline	11	9	5	6	4	5	n=40
Confounds & data quality	9	4	3	7	2	2	n=27
Scope & generalization	11	2	7	2	6	1	n=29
Quantitative reading	22	9	2	8	11	3	n=55
Core decision	9	4	2	3	3	2	n=23
Evidence integration	13	4	7	6	5	1	n=36
Experiment design	64	4			2		n=70

B

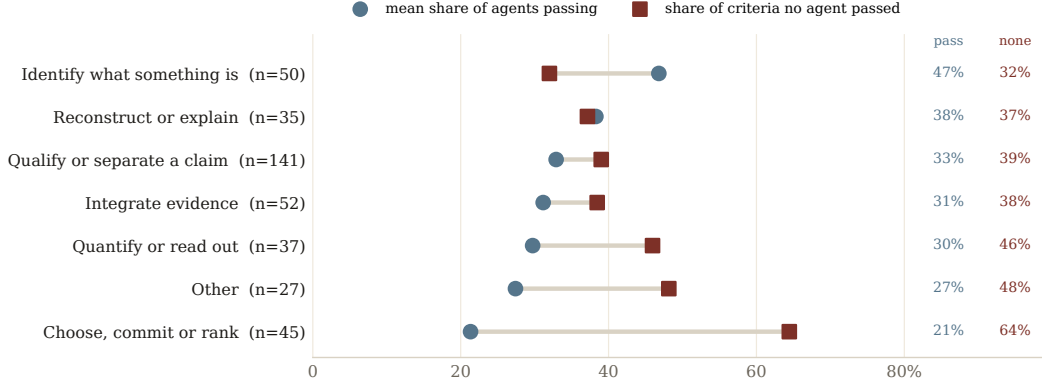


Figure 3: What agents pass and miss. **A**, Criteria passed by zero to five of the five agents, overall and by skill; shading is the share of the row, and the oxblood column is criteria no agent passed. **B**, By the move a criterion asks for (leading-verb heuristic): circles, mean share of agents passing; squares, share no agent passed. Criteria specifying the lab’s own planned experiment are omitted, since no agent chose it.

Known but unused. Two results show that agents hold the knowledge a missed decision needs.

- **Right move, wrong place.** In *t09*, Grok 4.7 correctly traced a displaced group of control wells to one plate, then designed its corrective experiment to exclude five other plates while keeping that one. In *m5-002*, all five agents’ code reproduced the lab’s regions exactly, yet seven edge-case criteria failed because no memo walked through the edge cases. In *m10-007*, four of five named the lab’s cell pair but not the evidence for it.
- **One sentence suffices.** For five never-passed core decisions, we appended one hint pointing at evidence the agent already had, without stating the answer; each hint shares at most one ordinary word with the PASS rule. GPT-6 Astra, which failed all five in the benchmark, passed all five (Figure 4).

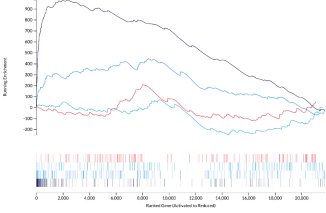
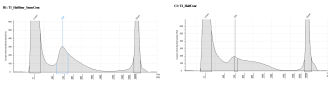
1. Evidence in the supplied records	2. Hint added to the brief	3. Criterion (core decision)	4. Change with the hint
m9-001 drug response, design  At 15 min, the compound's NF-κB signature is the strongest of four death inducers; at 1 h an inhibitor that rescues the cells lowers it about fourfold. No slide tests whether the response drives death.	“The excerpts show which pathways respond to [the compound], but not whether any of them contributes to death. Rank first the experiment that could tell a cause from a bystander.”	C-001. Experiment 1 must change NF-κB activity under the compound and measure both the pathway response and death, with causality still open.	Benchmark: 0/5 agents; four designed the NF-κB test but ranked it second. Hinted: PASS. “I prioritize direct NF-κB perturbation, followed by an HMOX1 knockout/complementation comparison.” Task 4/20 $\rightarrow$ 15/20.
t07 assay development, analyze  Left: 5 μL Tn5, 2 min (peak 210 bp, declining tail). Right: 2.5 μL Tn5, 5 min (peak 238 bp, broad large-fragment shoulder).	“Judge each tagmentation trace by its whole size distribution, not only by where its peak falls.”	C-003. Choose 5 μL Tn5 for 2 min, reasoning from the whole trace: the half-enzyme shoulder indicates incompletely cut DNA.	Benchmark: 0/5 agents; all chose the 238 bp peak. Hinted: PASS. “I favor the shorter reaction for its more compact whole distribution, not because 210 bp is an established optimum.”
nr-008 gene regulation, analyze Supplement: $S_{j,i} = \sum_k \mathbb{I}(p(mds_{j,i} > mds_{k,i}) < \alpha)$ counts one dataset’s wins against every other dataset; $A_{ct,i}$ then counts the <i>datasets</i> in a cell type that are significantly enriched.	“Check whether the evidence for a cell type could come mostly from a single dataset, and let that shape which analysis you choose.”	C-008. Choose repeated evidence across datasets, and explain that one unusually high dataset can supply many pairwise wins, which is why support is counted per dataset before per category.	Benchmark: 0/5 agents. Hinted: PASS. “One unusually high dataset can produce a long row/column of significant comparisons. Many such comparisons are not many independent supporting target datasets.” Task 7/20 $\rightarrow$ 12/20.
m3-010 nascent transcription, review The written analysis reports that 9% of unannotated transcribed regions contain both an open reading frame spanning at least 60% of the region and a bidirectional transcription start at its 5’ end, and that protein searches of this subgroup returned several hits.	“[The analysis] combined more than one line of evidence to select a subgroup. Decide what that combination supports for the subgroup, as distinct from all unannotated calls ...”	C-001. Commit to possible unannotated protein-coding genes for that subgroup, linking the ORF, the 5’ bidirectional signal and the protein matches, without claiming protein production.	Benchmark: 0/5 agents; four rejected any coding interpretation. Hinted: PASS. “The completed work supports a biologically motivated subgroup of candidate protein-coding transcripts ... but does not establish new protein-coding genes.”
m1-012 3D genome, review Figure 7: HMLα and HMRα sit 279 kb apart near opposite chromosome-3 ends, yet are $\sim$400 nm apart by FISH and 425 nm in the model.	“Decide whether the reported checks support any specific structural feature, such as how particular regions of a chromosome sit relative to each other.”	C-001. Endorse the ensemble for bounded interpretation, including loop-like proximity of chromosome ends, without certifying a unique fold.	Benchmark: 0/5 agents; four limited it to “coarse” geometry, one rejected it. Hinted: PASS. “Supports coarse juxtaposition of opposite end regions.”

Figure 4: *The decisions agents miss are within their reach.* Five core decisions that no benchmark agent passed. (1) Evidence the agent had. (2) The one sentence appended to the brief. (3) What the criterion requires. (4) Verdicts for the benchmark agents and the hinted run (GPT-6 Astra, same rubric), with a quote from the hinted submission. Bracketed text in hints is redacted; compound names and source text are withheld.

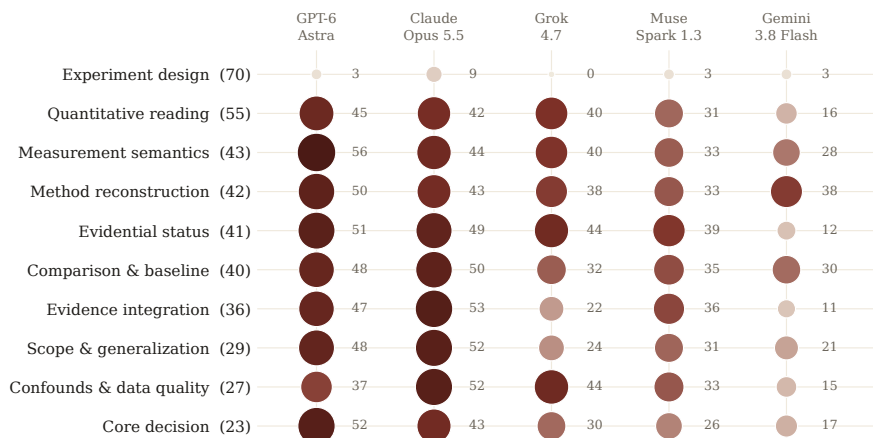
What the hints show. We present this probe to demonstrate that, in each of the five cases, the evidence provided could be interpreted to reach the correct conclusion. A single added sentence let the agent identify the correct approach by directing its attention to evidence it already had; no hint supplies domain knowledge (Figure 4).

- **m9-001:** the records show that a death-inducing compound triggers a stronger early NF- κ B signature than three other death inducers, and that an inhibitor which rescues the cells also lowers it, yet no experiment tests whether the response drives death. Four of five agents designed the functional NF- κ B test but ranked it second; a hint to separate cause from bystander moved it first and raised the task score from 4/20 to 15/20.
- **t07:** every agent chose the tagmentation condition with the larger 238 bp peak, overlooking its broad large-fragment shoulder. Asked to judge each trace by its whole size distribution, the agent chose the lab’s 5 μ L, 2-minute condition.
- **nr-008:** no agent explained that one unusually high dataset can supply many pairwise wins. A hint to check whether a cell type’s evidence came mostly from one dataset produced that explanation, and the task score rose from 7/20 to 12/20.
- **m3-010 and m1-012:** hints to decide what the combined evidence supports for a subgroup, and which specific structural features the checks support, produced the lab’s bounded conclusions: candidate protein-coding transcripts without a claim of protein production, and proximity of opposite chromosome ends without a unique fold.

We consider this exemplary evidence of a gap in reasoning rather than in knowledge: the agents hold the biological knowledge these decisions require, but do not bring it to bear unprompted.

Where agents differ. GPT-6 Astra leads on core decisions (52%) and measurement semantics (56%); Claude Opus 5.5 on evidence integration (53%), confounds and scope (52%) (Figure 5A). Gemini 3.8 Flash trails most on evidence integration and evidential status (11–12%). Assay development, target validation and hit-to-lead chemistry (25–28%) and reviews and designs (23–24%) are hardest (Figure 5B).

A



B

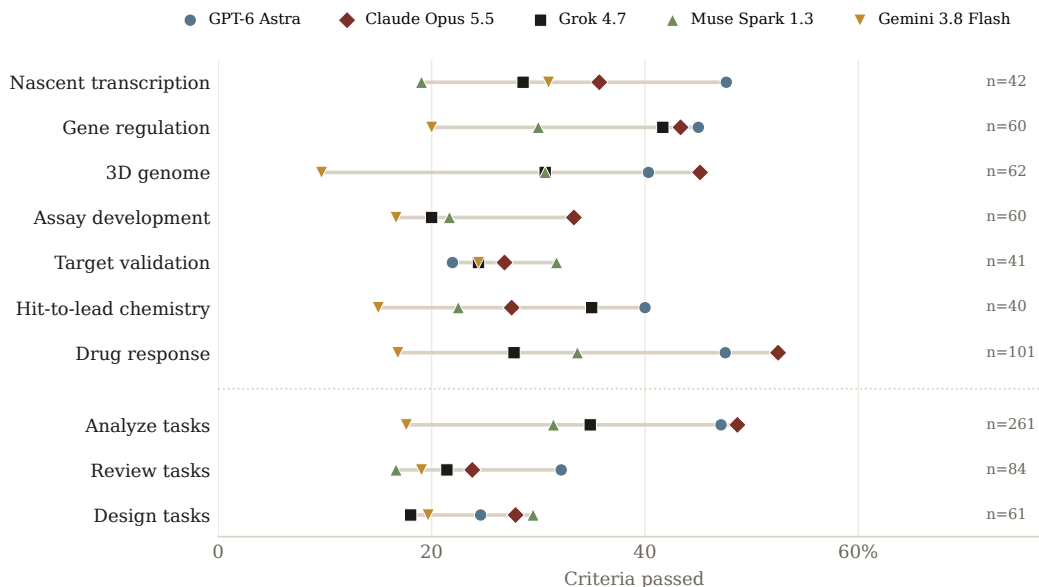


Figure 5: Where agents differ. **A**, Share of each skill’s criteria passed per agent (dot size and shade grow with the rate). **B**, Share passed per agent by theme and work type; n is the number of criteria.

5 Discussion

Our results demonstrate a clear reasoning bottleneck for frontier agents on real-world biological tasks. Their biological knowledge is frequently extensive, enough that agents can already work alongside expert biologists when experimental design stays with humans, but their ability to design experiments independently is weak. They often present several options, or otherwise fail to commit to an experiment and to discriminate the best one from the alternatives. In real biological experimentation, wall-clock time is irreducible: cells grow, differentiate and respond on their own schedules, and each experiment consumes weeks and material. As a result, autonomous and correct experimental design is arguably the single most important capability for making AI agents superhuman at these tasks.

LabBench measures this directly and penalizes failures along this axis. Improved results on

LabBench should indicate an improved ability for AI agents to carry out supervised, autonomous wet-lab experimentation, and eventually to run full programs themselves. To get there, models must develop a more cohesive world model of what experiments cost and of the specific evidence that would count against a hypothesis:

- **Choosing the discriminating experiment** (m10-002). A new assay for labeling nascent RNA joins a chemical handle to RNA ends; synthetic-oligo tests had shown that the join forms. Two untested steps could still break the assay, so the lab’s next two experiments isolated them cheaply: whether reverse transcriptase can copy across the chemical join, and whether the join forms in cell lysate. Four of five agents instead ranked first an experiment on polymerase incorporation of the handle, a step the lab had not flagged, and no agent placed the lysate test in its top two.
- **Weighing cost** (m10-009). The lab had already started a small pilot comparing a disease-variant cell line with its unedited parent, and planned a broad compound screen in the same pair only once the drug-free genotype signature proved repeatable. No agent committed to this cheap, gated path. Claude Opus 5.5 opened with a new differentiation of the edited lines into monocytes and macrophages and deferred the screen to a third stage; GPT-6 Astra began with a single-compound dose-response and then moved to 24-day differentiated cells. Both spend weeks of differentiation before establishing that the genotype signal exists.

6 Methods

Task format. Each task has a public brief and a private half. The brief states the question, names the deliverables the agent must write and lists the supplied documents. The private half holds the criteria and the hidden human records they were written from. Each criterion has a PASS rule, a FAIL rule, accepted alternatives, the visible evidence to use, the hidden record that supports it, an optional list of prerequisite criteria and a critical flag. LabBench v1 has 20 tasks and 406 criteria (20–22 per task; 30 critical, including all 23 core decisions; Appendix A).

Agents. Each agent ran in its vendor’s command-line harness: GPT-6 Astra in Codex, Claude Opus 5.5 in Claude Code at maximum effort, Grok 4.7 in Grok Build, Muse Spark 1.3 in Muse Code at high reasoning effort and Gemini 3.8 Flash in Gemini CLI, with other settings at harness defaults. Every run started on a fresh virtual machine with the brief as its prompt and the documents in its workspace, had no human interaction and was limited to about five hours; network access was not restricted. Each agent ran once per task (100 runs).

Grading. A judge (Codex with gpt-5.6-sol at extra-high reasoning effort) grades each run in a fresh session. It sees the brief, the supplied documents, the private half of the task and the agent’s deliverables, but not the agent’s trace. It judges every criterion once, returning PASS or FAIL with a reason and citations to the submission, the input documents and the human evidence. When a criterion’s rules make it conditional on an earlier conclusion, it fails if that conclusion is wrong or missing, and a missing deliverable fails every criterion that depends on it. A task’s score is the share of its criteria passed; critical criteria are not weighted.

Analysis. Model scores are unweighted means over tasks, with 95% percentile bootstrap intervals from 5,000 resamples of tasks. Agreement between two agents is the share of the 406 criteria on which their verdicts match; Cohen’s κ [20] corrects it for chance using each agent’s overall pass rate. Each criterion carries one primary skill from Table 2, assigned by an AI annotator (Claude Opus

5.5) from the criterion’s title, PASS rule and hidden evidence; the labels were not independently double-coded. Figure 3B groups criteria by the leading verb of their title.

Hint probe. We chose five core decisions that no agent passed, that have no prerequisite criteria, and that come from five different themes. For each, one sentence was written after reading the failed judgments. It points at evidence already in the packet without stating the answer, and shares at most one content word with the criterion’s PASS rule that the brief lacks. The hinted task is the published task with that sentence appended to the brief. GPT-6 Astra solved each hinted task once, and gpt-5.6-sol graded it against the unchanged criteria. The comparison is GPT-6 Astra’s benchmark run on the same task.

References

- [1] Hendrycks, D. et al. Measuring Massive Multitask Language Understanding. ICLR 2021. arXiv:2009.03300.
- [2] Lu, P. et al. Learn to Explain: Multimodal Reasoning via Thought Chains for Science Question Answering. NeurIPS 2022. arXiv:2209.09513.
- [3] Rein, D. et al. GPQA: A Graduate-Level Google-Proof Q&A Benchmark. arXiv:2311.12022, 2023.
- [4] Phan, L. et al. Humanity’s Last Exam. arXiv:2501.14249, 2025.
- [5] Laurent, J. M. et al. LAB-Bench: Measuring Capabilities of Language Models for Biology Research. arXiv:2407.10362, 2024.
- [6] O’Donoghue, O. et al. BioPlanner: Automatic Evaluation of LLMs on Protocol Planning in Biology. EMNLP 2023. arXiv:2310.10632.
- [7] Chen, Z. et al. ScienceAgentBench: Toward Rigorous Assessment of Language Agents for Data-Driven Scientific Discovery. ICLR 2025. arXiv:2410.05080.
- [8] Majumder, B. P. et al. DiscoveryBench: Towards Data-Driven Discovery with Large Language Models. arXiv:2407.01725, 2024.
- [9] Mitchener, L. et al. BixBench: A Comprehensive Benchmark for LLM-based Agents in Computational Biology. arXiv:2503.00096, 2025.
- [10] Siegel, Z. S. et al. CORE-Bench: Fostering the Credibility of Published Research Through a Computational Reproducibility Agent Benchmark. arXiv:2409.11363, 2024.
- [11] Tian, M. et al. SciCode: A Research Coding Benchmark Curated by Scientists. NeurIPS 2024 Datasets and Benchmarks. arXiv:2407.13168.
- [12] Starace, G. et al. PaperBench: Evaluating AI’s Ability to Replicate AI Research. arXiv:2504.01848, 2025.
- [13] Arora, R. K. et al. HealthBench: Evaluating Large Language Models Towards Improved Human Health. arXiv:2505.08775, 2025.
- [14] Harvey AI. Harvey LAB: The Legal Agent Benchmark, v1.0. <https://github.com/harveyai/harvey-labs/tree/v1.0>, 2026. Announcement: <https://www.harvey.ai/blog/introducing-harveys-legal-agent-benchmark>.
- [15] Zheng, L. et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. NeurIPS 2023. arXiv:2306.05685.

- [16] Wei, J. et al. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. NeurIPS 2022. arXiv:2201.11903.
- [17] Kojima, T. et al. Large Language Models are Zero-Shot Reasoners. NeurIPS 2022. arXiv:2205.11916.
- [18] Phuong, M. et al. Evaluating Frontier Models for Dangerous Capabilities. arXiv:2403.13793, 2024.
- [19] Shi, F. et al. Large Language Models Can Be Easily Distracted by Irrelevant Context. ICML 2023. arXiv:2302.00093.
- [20] Cohen, J. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* 20(1):37–46, 1960.

A The 20 tasks

Task	Title	Theme	Type	Crit.	Astra	Opus	Grok	Muse	Gemini
m6-004	Compare the chromatin context of early- and late-bound transcription-factor sites	Gene regulation	analyze	20	50	45	35	35	30
m5-002	Reconstruct how control genomic regions were chosen in a nascent-RNA analysis	Nascent transcription	analyze	20	45	40	30	25	55
m9-001	Choose the next two experiments for a compound-induced transcriptional response	Drug-response profiling	design	20	20	20	15	20	20
m1-012	Assess a 3D chromosome model against imaging distance measurements	3D genome	review	22	23	23	14	14	9
t09	Separate plate artifacts from drug effects in a transcriptomic screen	Drug-response profiling	analyze	21	24	67	38	33	10
m1-005	Interpret cell-type differences in 3D folding of a gene locus	3D genome	analyze	20	45	55	45	25	10
m10-007	Choose a matched cell comparison before testing a DNA-repair vulnerability	Target & disease-model validation	review	20	20	25	25	25	40
m9-010	Assess what an untreated pilot of a nascent-transcription assay establishes	Assay & library development	analyze	20	35	50	30	25	10
t07	Choose library preparation conditions from fragment-size traces	Assay & library development	analyze	20	35	15	15	10	10
m10-009	Choose the next screen in a disease-variant cell model	Target & disease-model validation	design	21	24	29	24	38	10
m10-002	Choose the next validation tests for a nascent-RNA labeling assay	Assay & library development	design	20	30	35	15	30	30
m10-003	Weigh transcriptional and binding evidence for two pathway-activating hits	Hit-to-lead chemistry	analyze	20	45	40	50	30	10
nr-008	Decide what motif-enrichment differences show about cell-type activity	Gene regulation	analyze	20	35	40	45	20	20
m8-001	Choose provisional transcriptional markers of target inhibition	Drug-response profiling	analyze	20	85	95	0	30	40
t15	Reassess a compound series and its lead after new data	Hit-to-lead chemistry	review	20	35	15	20	15	20
m3-010	Assess candidate protein-coding transcripts outside gene annotations	Nascent transcription	review	22	50	32	27	14	9

Task	Title	Theme	Type	Crit.	Astra	Opus	Grok	Muse	Gemini
m1-006	Reconcile robustness and evolvability of transcription-factor binding sites	Gene regulation	analyze	20	50	45	45	35	10
m6-002	Reconstruct the comparison behind a drug-resistance transcriptional analysis	Drug-response profiling	analyze	20	55	25	15	30	5
m1-008	Choose a model for interpreting the 3D structure of a gene cluster	3D genome	analyze	20	55	60	35	55	10
m8-003	Choose a transcriptional signature for a drug-combination response	Drug-response profiling	analyze	20	55	55	70	55	10

Scores are percentages of criteria passed.